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A Experimental Prompts

To facilitate robust and reproducible semantic analysis of RLM-generated reasoning traces, we design two explicit prompting templates, detailed documented in Template 9 and Template 10.

Template 9 guides the model to cluster reasoning units into coherent, higher-level reasoning steps. This step provides context-aware candidate instances of clustering of long chain-of-thought tokens.

Template 10 is used to extract the semantic relationship between every pair of reasoning steps. By explicitly labeling each pair as `support`, `contradict`, or `independent`, this template enables the later probabilistic estimation of interdependencies within the model's chain-of-thought.

B Few-Shot Prompting Styles

Most existing research works reporting the performance degradation of RLM given few-shot prompting have not explicitly analyzed the structure of few-shot examples, while the concrete formulation of few-shot demonstrations could play a significant role on the behavior of language model. To isolate potential structural factors contributing to this performance degradation, in this paper, we introduce and analyze three distinct few-shot prompting style:

i) `Minimal:` Minimal exemplars containing only problem statements and final answers, without intermediate reasoning steps or explanatory content.

ii) `Concise:` Human-authored concise reasoning traces characterized by *short*, linear progression from problem formulation to solution with minimal exploration.

iii) `Explanatory:` RLM-generated *long* reasoning sequences featuring extensive problem space exploration, iterative verification mechanisms, and explicit self-reflection.

C Implementation Details

Models We evaluate a range of reasoning LLMs, including DeepSeek-R1-distilled-Llama-8B, DeepSeek-R1-distilled-Qwen-14B, and QWEN3-32B (Yang et al., 2025a). For conditional sampling (P_{LLM}), we adopt DEEPSEEK-V3-0324 (DeepSeek-AI et al., 2024).

Hyper-parameters To ensure reproducibility, we set the generation temperature to 0 when producing reasoning chains (CoT) with RLMs. During logical clustering and semantics detection processes, we keep $\tau_r \sim [0.3, 0.7]$ to ensure sampling diversity.

Datasets All experiments are conducted on the GPQA-Diamond benchmark (Rein et al., 2023). To facilitate robust reasoning analysis and avoid potential training data contamination, we convert multiple-choice items into open-ended questions, requiring models to actively generate reasoning CoT as well as final answer rather than matching existing choices.

$\mathcal{P}_{\text{clu}}$ Logical Clustering

Instruction:
You are given a sequence of reasoning units, each representing a contiguous fragment from a language model's chain-of-thought (CoT) output. These units have typically been segmented using raw delimiters and may be overly fine-grained or fragmented for downstream analysis.
[*Input Template*]

Your task is to cluster consecutive reasoning units that are semantically connected, producing a concise and coherent set of higher-level reasoning steps. Each reasoning step should: - Combine all units that express a single coherent sub-task, logical inference, or closely related set of thoughts. Aim to group together units that collectively advance the same intermediate goal or logical point. - Ensure that each resulting reasoning step contains enough self-contained context to be analyzed independently, but avoid excessive merging that would result in overly broad or incoherent segments. - Maintain the original sequential order of reasoning. - Avoid splitting apart reasoning units that clearly belong to the same sub-problem or share strong contextual dependency. - Use concise yet informative titles for each reasoning step, reflecting its main logical function or purpose (e.g., "Restate Problem", "Recall Known Facts", "Solve Equation", "Synthesize Solution", etc.).

Expected Output Format:

```
{
  "s0": {"title": "...", "content": "..."},
  "s1": {"title": "...", "content": "..."},
  ...
}
```

where each `"sX"` key indexes an ordered reasoning step, with an appropriate `"title"` summarizing its logical purpose and `"content"` containing the merged, cleaned reasoning text.

Please ensure the output is structured, coherent, and well-suited for subsequent semantic analysis or graph-based modeling of the reasoning process.

Output: {"s_0": {"title":..., "content":...},...}

Figure 9: Complete instruction ($\mathcal{P}_{\text{clu}}$) for clustering reasoning units into logical cohesive reasoning steps.

D algorithm

This appendix provides the detailed pseudocode for the core algorithmic components of our framework:

$\mathcal{P}_{\text{sem}}$ Semantics Detection

Instruction:
Given an ordered sequence of K reasoning steps, each representing a semantically meaningful stage in a language model's chain-of-thought output, your task is to systematically assess the semantic relationship between each pair of reasoning steps.
[*Logical Steps Template*]

For every ordered pair (i, j) with $1 \leq i < j \leq K$, your task is to decide whether step i: - `supports` step j (i.e., provides information, justification, intermediate results, or logical basis for step j), - `contradicts` step j (i.e., conflicts with, undermines, or provides an incompatible claim or result relative to step j), or - is `independent` of step j (i.e., is neither directly supportive nor contradictory; the steps are unrelated in logical content).
When making your decision, consider both explicit logical connections (e.g., mathematical derivation, use of previous results, direct contradiction) and more implicit semantic dependencies (e.g., fact recall enabling downstream calculations).

Expected Output Format:

```
{
  "(0,1)": "support",
  "(0,2)": "independent",
  "(1,2)": "contradict",
  ...
}
```

where each key `"(i,j)"` denotes an ordered pair of step indices (with $i < j$), and each value is one of `"support"`, `"contradict"`, or `"independent"`.

For each pair, provide only one label reflecting the most salient semantic relationship. If the relationship is unclear or borderline, default to `independent` unless clear evidence suggests support or contradiction.

Ensure that all pairs (i, j) with $1 \leq i < j \leq K$ are covered in the output, and that your decisions are consistent and justifiable based on the provided reasoning steps.

Output: {(S_0, S_1):..., (S_0, S_2): ..., (S_1, S_0): ...}

Figure 10: Complete instruction ($\mathcal{P}_{\text{sem}}$) for detecting semantical relationship among all reasoning steps.

ensemble-based clustering of reasoning units 1, and adaptive sampling-based construction of the semantic dependency graph 2. These algorithms operationalize the methods described in the main text, clarifying the iterative processes and statistical aggregation techniques used to ensure robust, uncertainty-aware structure induction from RLM outputs.

E Reference Step Length μ_{ref}

The ideal average step length μ_{ref} serves as a reference for the length-regularity term and is computed by dividing the total number of tokens N by a target number of reasoning steps K_{target}. We

Algorithm 1: Ensemble-Based Clustering of Reasoning Units

Input: Reasoning units U, clustering prompt $\mathcal{P}_{\text{cluster}}$, number of samples B, temperature grid $\{\tau_1, \ldots, \tau_B\}$
Output: Selected segmentation C^*

```
for b ← 1 to B do
    C^(b) ~ P_LLM(S | P_cluster, U; τ_b);
    F^(b) ← F(C^(b));
C* ← arg max_b F^(b);
return C*;
```

set $K_{\text{target}} = \min(\max(3, \lceil\sqrt{M}\rceil), 30)$, where M is the number of initial delimiter-based segments. This square-root heuristic with lower and upper bounds ensures μ_ref adapts to different CoT lengths and discourages both over-segmentation and overly coarse steps, enabling a scale-invariant and task-agnostic regularization.

F Standard-Error Derivation for the Signed Edge Confidence

Let a single LLM adjacency sample yield a label $c \in \{-1, 0, +1\}$ for the ordered pair (i, j), corresponding to `contradict`, `independent`, and `support`, respectively. Define the random variable

$$Z = \begin{cases} +1, & c = +1, \\ -1, & c = -1, \\ 0, & c = 0. \end{cases}$$

The *signed edge confidence* is the empirical mean of Z over R samples,

$$w_{ij} = \frac{1}{R}\sum_{r=1}^{R} Z^{(r)} = \hat{p}_{ij}(+1) - \hat{p}_{ij}(-1),$$

where $\hat{p}_{ij}(c) = \frac{1}{R}\sum_r \mathbf{1}[c^{(r)} = c]$.

Because $\mathbb{E}[Z] = p(+1) - p(-1)$, only the two informative labels contribute to the signed mean. The variance of Z under the true distribution p is

$$\begin{aligned} &\mathrm{Var}(Z) \\ &= (+1)^2 p(+1) + (-1)^2 p(-1) + 0^2 p(0) \\ &- \big[p(+1) - p(-1)\big]^2 \\ &= p(+1) + p(-1) - \big[p(+1) - p(-1)\big]^2. \end{aligned}$$

Replacing $p(\cdot)$ with their empirical estimates and dividing by R yields an unbiased standard-error